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$$\int \frac{2x\left(x+5\right)}{x^4-1}dx = \int \frac{2x^2+10x}{\left(x^2-1\right)\left(x^2+1\right)}dx = \int \frac{2x^2+10x}{\left(x-1\right)\left(x+1\right)\left(x^2+1\right)}dx$$

$$\rightarrow \frac{2x^2+10x}{\left(x-1\right)\left(x+1\right)\left(x^2+1\right)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$$

$$\Rightarrow A = \frac{2+10}{\left(1+1\right)\left(1+1\right)} = \frac{12}{4} = 3$$

$$\Rightarrow B = \frac{2-10}{\left(-1-1\right)\left(1+1\right)} = \frac{-8}{-4} = 2$$

$$\Rightarrow C\left(i\right) + D = \frac{-2+10i}{\left(i-1\right)\left(i+1\right)} = \frac{10i-2}{-2} = 1-5i$$

$$\Rightarrow C = -5, D = 1$$

$$= 3\int \frac{1}{x-1}dx + 2\int \frac{1}{x+1} + \int \frac{-5x+1}{x^2+1}dx$$

$$= 3\ln|x-1| + 2\ln|x+1| - \frac{5}{2}\int \frac{1}{x^2+1}d\left(x^2+1\right) + \int \frac{1}{x^2+1}dx$$

$$= 3\ln|x-1| + 2\ln|x+1| - \frac{5}{2}\ln|x^2+1| + \operatorname{Bgtan} x + c$$

References